

former category, we include so-called *descriptive statistics* that characterize a data set in general terms: its mean, variance, and so on. We also include statistical tests that seek to establish the “sameness” or “differentness” of two or more data sets, or that seek to establish and measure a degree of *correlation* between two data sets. These subjects are discussed in this chapter.

In the other category, model-dependent statistics, we lump the whole subject of fitting data to a theory, parameter estimation, least-squares fits, and so on. Those subjects are introduced in Chapter 15.

Section 14.1 deals with so-called *measures of central tendency*, the moments of a distribution, the median and mode. In §14.2 we learn to test whether different data sets are drawn from distributions with different values of these measures of central tendency. This leads naturally, in §14.3, to the more general question of whether two distributions can be shown to be (significantly) different.

In §14.4 – §14.7, we deal with *measures of association* for two distributions. We want to determine whether two variables are “correlated” or “dependent” on one another. If they are, we want to characterize the degree of correlation in some simple ways. The distinction between parametric and nonparametric (rank) methods is emphasized. Information-theoretic methods are discussed in §14.7. Section 14.9 introduces the concept of data smoothing, and discusses the particular case of Savitzky-Golay smoothing filters.

This chapter draws mathematically on the material on special functions that was presented in Chapter 6, especially §6.1 – §6.4 and §6.14. You may wish, at this point, to review those sections.

Bayesian methods make little appearance in this chapter, but become more prominent in the two chapters following this one.

CITED REFERENCES AND FURTHER READING:

- Bevington, P.R., and Robinson, D.K. 2002, *Data Reduction and Error Analysis for the Physical Sciences*, 3rd ed. (New York: McGraw-Hill).
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- Devore, J.L. 2003, *Probability and Statistics for Engineering and the Sciences*, 6th ed. (Belmont, CA: Duxbury Press).
- Wall, J.V., and Jenkins, C.R. 2003, *Practical Statistics for Astronomers* (Cambridge, UK: Cambridge University Press).
- Lupton, R. 1993, *Statistics in Theory and Practice* (Princeton, NJ: Princeton University Press).

14.1 Moments of a Distribution: Mean, Variance, Skewness, and So Forth

When a set of values has a sufficiently strong central tendency, that is, a tendency to cluster around some particular value, then it may be useful to characterize the set by a few numbers that are related to its *moments*, the sums of integer powers of the values.

Best known is the *mean* of the values $x_0, \dots, x_{N-1}$,

$$\bar{x} = \frac{1}{N} \sum_{j=0}^{N-1} x_j \quad (14.1.1)$$

which estimates the value around which central clustering occurs. Note the use of an overbar to denote the mean; angle brackets are an equally common notation, e.g., $\langle x \rangle$. You should be aware that the mean is not the only available estimator of this quantity, nor is it necessarily the best one. For values drawn from a probability distribution with very broad “tails,” the mean may converge poorly, or not at all, as the number of sampled points is increased. Alternative estimators, the *median* and the *mode*, are mentioned at the end of this section.

Having characterized a distribution’s central value, one conventionally next characterizes its “width” or “variability” around that value. Here again, more than one measure is available. Most common is the *variance*,

$$\text{Var}(x_0 \dots x_{N-1}) = \frac{1}{N-1} \sum_{j=0}^{N-1} (x_j - \bar{x})^2 \quad (14.1.2)$$

or its square root, the *standard deviation*,

$$\sigma(x_0 \dots x_{N-1}) = \sqrt{\text{Var}(x_0 \dots x_{N-1})} \quad (14.1.3)$$

Equation (14.1.2) estimates the mean squared deviation of x from its mean value. There is a long story about why the denominator of (14.1.2) is $N-1$ instead of N . If you have never heard that story, you should consult any good statistics text. Here we will be content to note that the $N-1$ *should* be changed to N if you are ever in the situation of measuring the variance of a distribution whose mean $\bar{x}$ is known a priori rather than being estimated from the data. (We might also comment that if the difference between N and $N-1$ ever matters to you, then you are probably up to no good anyway — e.g., trying to substantiate a questionable hypothesis with marginal data.)

If we calculate equation (14.1.1) many times with different sets of sampled data (each set having N values), the values $\bar{x}$ will themselves have a standard deviation. This is called the *standard error* of the estimated mean $\bar{x}$. When the underlying distribution is Gaussian, it is given approximately by $\sigma/\sqrt{N}$. Correspondingly, there is a standard error of the estimated variance, equation (14.1.2), which is approximately $\sigma^2\sqrt{2/N}$, and a standard error for the estimated σ , equation (14.1.3), which is approximately $\sigma/\sqrt{2N}$.

As the mean depends on the first moment of the data, so do the variance and standard deviation depend on the second moment. It is not uncommon, in real life, to be dealing with a distribution whose second moment does not exist (i.e., is infinite). In this case, the variance or standard deviation is useless as a measure of the data’s width around its central value: The values obtained from equations (14.1.2) or (14.1.3) will not converge with increased numbers of points, nor show any consistency from data set to data set drawn from the same distribution. This can occur even when the width of the peak looks, by eye, perfectly finite. A more robust estimator

of the width is the *average deviation* or *mean absolute deviation*, defined by

$$\text{ADev}(x_0 \dots x_{N-1}) = \frac{1}{N} \sum_{j=0}^{N-1} |x_j - \bar{x}| \quad (14.1.4)$$

One often substitutes the sample median x_{med} for $\bar{x}$ in equation (14.1.4). For any fixed sample, the median in fact minimizes the mean absolute deviation.

Statisticians have historically sniffed at the use of (14.1.4) instead of (14.1.2), since the absolute value brackets in (14.1.4) are “nonanalytic” and make theorem-proving more difficult. In recent years, however, the fashion has changed, and the subject of *robust estimation* (meaning, estimation for broad distributions with significant numbers of “outlier” points) has become a popular and important one. Higher moments, or statistics involving higher powers of the input data, are almost always less robust than lower moments or statistics that involve only linear sums or (the lowest moment of all) counting.

That being the case, the *skewness* or *third moment*, and the *kurtosis* or *fourth moment* should be used with caution or, better yet, not at all.

The skewness characterizes the degree of asymmetry of a distribution around its mean. While the mean, standard deviation, and average deviation are *dimensional* quantities, that is, have the same units as the measured quantities x_j , the skewness is conventionally defined in such a way as to make it *nondimensional*. It is a pure number that characterizes only the shape of the distribution. The usual definition is

$$\text{Skew}(x_0 \dots x_{N-1}) = \frac{1}{N} \sum_{j=0}^{N-1} \left[\frac{x_j - \bar{x}}{\sigma} \right]^3 \quad (14.1.5)$$

where $\sigma = \sigma(x_0 \dots x_{N-1})$ is the distribution’s standard deviation (14.1.3). A positive value of skewness signifies a distribution with an asymmetric tail extending out toward more positive x ; a negative value signifies a distribution whose tail extends out toward more negative x (see Figure 14.1.1).

Of course, any set of N measured values is likely to give a nonzero value for (14.1.5), even if the underlying distribution is in fact symmetrical (has zero skewness). For (14.1.5) to be meaningful, we need to have some idea of *its* standard error. Unfortunately, that depends on the shape of the underlying distribution, and rather critically on its tails! For the idealized case of a normal (Gaussian) distribution, the standard error of (14.1.5) is approximately $\sqrt{15/N}$ when $\bar{x}$ is the true mean and $\sqrt{6/N}$ when it is estimated by the sample mean, (14.1.1). (Yes, using the sample mean is likely to give a more accurate estimate than using the true mean!) In real life it is good practice to believe in skewnesses only when they are several or many times as large as this.

The kurtosis is also a nondimensional quantity. It measures the relative peakedness or flatness of a distribution. Relative to what? A normal distribution! What else? A distribution with positive kurtosis is termed *leptokurtic*; the outline of the Matterhorn is an example. A distribution with negative kurtosis is termed *platykurtic*; the outline of a loaf of bread is an example. (See Figure 14.1.1.) And, as you no doubt expect, an in-between distribution is termed *mesokurtic*.

The conventional definition of the kurtosis is

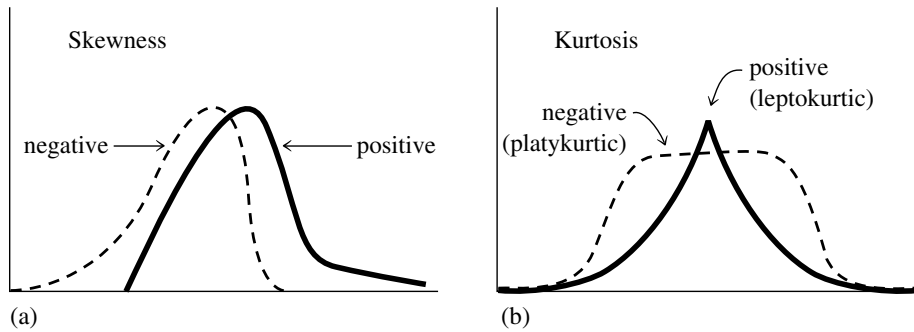


Figure 14.1.1. Distributions whose third and fourth moments are significantly different from a normal (Gaussian) distribution. (a) Skewness or third moment. (b) Kurtosis or fourth moment.

$$\text{Kurt}(x_0 \dots x_{N-1}) = \left\{ \frac{1}{N} \sum_{j=0}^{N-1} \left[\frac{x_j - \bar{x}}{\sigma} \right]^4 \right\} - 3 \quad (14.1.6)$$

where the -3 term makes the value zero for a normal distribution.

The standard error of (14.1.6) as an estimator of the kurtosis of an underlying normal distribution is $\sqrt{96/N}$ when σ is the true standard deviation, and $\sqrt{24/N}$ when it is the sample estimate (14.1.3). (Yes, you are better off using the sample variance.) However, the kurtosis depends on such a high moment that there are many real-life distributions for which the standard deviation of (14.1.6) as an estimator is effectively infinite.

Calculation of the quantities defined in this section is perfectly straightforward. Many textbooks use the binomial theorem to expand out the definitions into sums of various powers of the data, e.g., the familiar

$$\text{Var}(x_0 \dots x_{N-1}) = \frac{1}{N-1} \left[\left(\sum_{j=0}^{N-1} x_j^2 \right) - N\bar{x}^2 \right] \approx \bar{x}^2 - \bar{x}^2 \quad (14.1.7)$$

but this can magnify the roundoff error by a large factor and is generally unjustifiable in terms of computing speed. A clever way to minimize roundoff error, especially for large samples, is to use the *corrected two-pass algorithm* [1]: First calculate $\bar{x}$, then calculate $\text{Var}(x_0 \dots x_{N-1})$ by

$$\text{Var}(x_0 \dots x_{N-1}) = \frac{1}{N-1} \left\{ \sum_{j=0}^{N-1} (x_j - \bar{x})^2 - \frac{1}{N} \left[\sum_{j=0}^{N-1} (x_j - \bar{x}) \right]^2 \right\} \quad (14.1.8)$$

The second sum would be zero if $\bar{x}$ were exact, but otherwise it does a good job of correcting the roundoff error in the first term.

moment.h

```
void moment(VecDoub_I &data, Doub &ave, Doub &adev, Doub &sdev, Doub &var,
Doub &skew, Doub &curt) {
```

Given an array of data[0..n-1], this routine returns its mean ave, average deviation adev, standard deviation sdev, variance var, skewness skew, and kurtosis curt.

```
Int j,n=data.size();
Doub ep=0.0,s,p;
```

```

if (n <= 1) throw("n must be at least 2 in moment");
s=0.0;
for (j=0;j<n;j++) s += data[j];
ave=s/n;
adev=var=skew=curt=0.0;
for (j=0;j<n;j++) {
    adev += abs(s=data[j]-ave);
    ep += s;
    var += (p=s*s);
    skew += (p *= s);
    curt += (p *= s);
}
adev /= n;
var=(var-ep*ep/n)/(n-1);
sdev=sqrt(var);
if (var != 0.0) {
    skew /= (n*var*sdev);
    curt=curt/(n*var*var)-3.0;
} else throw("No skew/kurtosis when variance = 0 (in moment)");
}

```

First pass to get the mean.

Second pass to get the first (absolute), second, third, and fourth moments of the deviation from the mean.

Corrected two-pass formula.
Put the pieces together according to the conventional definitions.

14.1.1 Semi-Invariants

The mean and variance of independent random variables are additive: If x and y are drawn independently from two, possibly different, probability distributions, then

$$\overline{(x+y)} = \bar{x} + \bar{y} \quad \text{Var}(x+y) = \text{Var}(x) + \text{Var}(y) \quad (14.1.9)$$

Higher moments are not, in general, additive. However, certain combinations of them, called *semi-invariants*, are in fact additive. If the centered moments of a distribution are denoted M_k ,

$$M_k \equiv \left\langle (x_i - \bar{x})^k \right\rangle \quad (14.1.10)$$

so that, e.g., $M_2 = \text{Var}(x)$, then the first few semi-invariants, denoted I_k , are given by

$$\begin{aligned} I_2 &= M_2 & I_3 &= M_3 & I_4 &= M_4 - 3M_2^2 \\ I_5 &= M_5 - 10M_2M_3 & I_6 &= M_6 - 15M_2M_4 - 10M_3^2 + 30M_2^3 \end{aligned} \quad (14.1.11)$$

Notice that the skewness and kurtosis, equations (14.1.5) and (14.1.6), are simple powers of the semi-invariants,

$$\text{Skew}(x) = I_3/I_2^{3/2} \quad \text{Kurt}(x) = I_4/I_2^2 \quad (14.1.12)$$

A Gaussian distribution has all its semi-invariants higher than I_2 equal to zero. A Poisson distribution has all of its semi-invariants equal to its mean. For more details, see [2].

14.1.2 Median and Mode

The median of a probability distribution function $p(x)$ is the value x_{med} for which larger and smaller values of x are equally probable:

$$\int_{-\infty}^{x_{\text{med}}} p(x) dx = \frac{1}{2} = \int_{x_{\text{med}}}^{\infty} p(x) dx \quad (14.1.13)$$

The median of a distribution is estimated from a sample of values $x_0, \dots, x_{N-1}$ by finding that value x_i which has equal numbers of values above it and below it. Of course, this is not possible when N is even. In that case it is conventional to